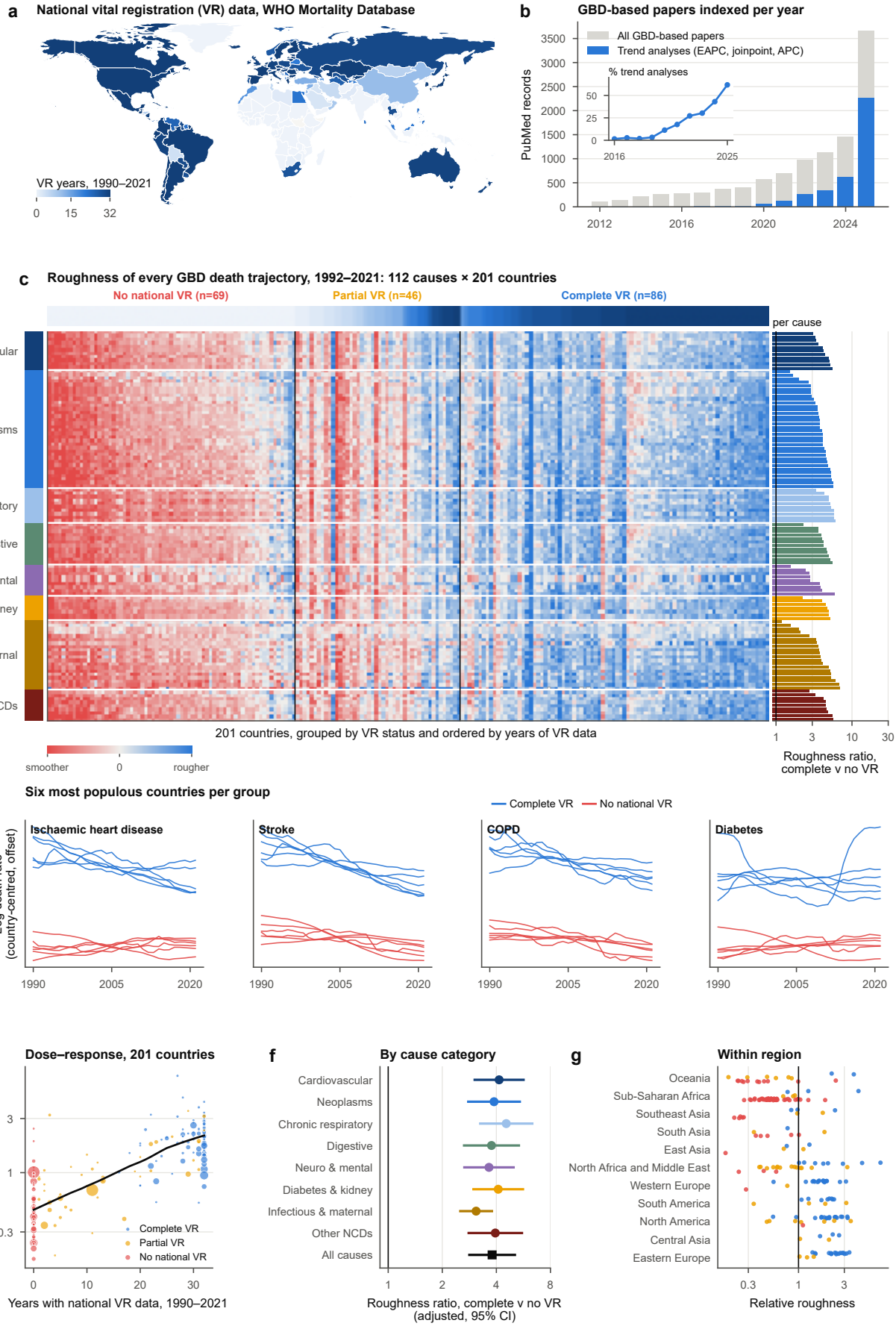

a
National vital registration (VR) data, WHO Mortality Database
VR years, 1990–2021
0
15
32
b
GBD-based papers indexed per year
All GBD-based papers
Trend analyses (EAPC, joinpoint, APC)
% trend analyses
PubMed records
2012
2016
2020
2024
c
Roughness of every GBD death trajectory, 1992–2021: 112 causes × 201 countries
No national VR (n=69)
Partial VR (n=46)
Complete VR (n=86)
per cause
Cardiovascular
Neoplasms
Chronic respiratory
Digestive
Neuro & mental
Diabetes & kidney
Infectious & maternal
Other NCDs
201 countries, grouped by VR status and ordered by years of VR data
smoother
0
rougher
Roughness ratio, complete v no VR
d
Six most populous countries per group
Complete VR
No national VR
Ischaemic heart disease
Stroke
COPD
Diabetes
Log death rate (country-centred, offset)
e
Dose–response, 201 countries
Relative roughness
Complete VR
Partial VR
No national VR
Years with national VR data, 1990–2021
f
By cause category
Cardiovascular
Neoplasms
Chronic respiratory
Digestive
Neuro & mental
Diabetes & kidney
Infectious & maternal
Other NCDs
All causes
Roughness ratio, complete v no VR (adjusted, 95% CI)
g
Within region
Oceania
Sub-Saharan Africa
Southeast Asia
South Asia
East Asia
North Africa and Middle East
Western Europe
South America
North America
Central Asia
Eastern Europe
Relative roughness